Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line: n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

We see 5 - 1 = 4

So Return 1.

For example:

Input:	Result
4 1 2 4 5 3	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n, k;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &k);

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[j] - a[i] == k) {
                printf("1");
                return 0;
            }
        }
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
✔	4 1 2 4 5 3	1	1	✔
✔	10 1 3 6 8 12 15 16 20 21 24 1	1	1	✔
✔	10 1 2 3 5 10 15 16 21 28 29 0	0	0	✔
✔	10 0 2 3 7 12 15 16 20 21 24 10	1	1	✔

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k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase

YES as 5 - 1 = 4

So Return 1.

For example:

Input:	Result:
4 1 2 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int n, k;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &k);
    int i = 0, j = 1;
    while (j < n) {
        if (a[j] - a[i] == k) {
            printf("1");
            return 0;
        }
        else if (a[j] - a[i] < k)
            j++;
        else
            i++;
        if (i == j)
            j++;
    }
    printf("0");
    return 0;
}
```

	Input	Expected	Got	
✔	3 5 0 5 0	1	1	✔
✔	10 1 2 4 8 12 37 15 38 21 24 3	1	1	✔
✔	10 1 2 4 5 15 37 16 21 28 29 0	0	0	✔
✔	10 0 2 4 7 13 37 15 38 21 25 38	1	1	✔

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 5 6

Output:

5 6

For example:

Input	Result
1 4 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int T;
    scanf("%d", &T);
    while (T--) {
        int n1, n2;
        scanf("%d", &n1);
        int a[n1];
        for (int i = 0; i < n1; i++)
            scanf("%d", &a[i]);
        scanf("%d", &n2);
        int b[n2];
        for (int i = 0; i < n2; i++)
            scanf("%d", &b[i]);
        int i = 0, j = 0;
        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✓	1 4 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 3 2 5 6	5 6	5 6	✓

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Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 248

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 5 6

Output:

5 6

For example:

Input	Result
1 4 10 17 57 6 2 7 10 15 57 248	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2;
        scanf("%d", &n1);

        int a[n1];
        for (int i = 0; i < n1; i++)
            scanf("%d", &a[i]);

        scanf("%d", &n2);

        int b[n2];
        for (int i = 0; i < n2; i++)
            scanf("%d", &b[i]);

        int i = 0, j = 0;

        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j])
                i++;
            else
                j++;
        }

        printf("\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 4 10 17 57 6 2 7 10 15 57 248	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 5 6	5 6	5 6	✔

Find Duplicate in Array

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element a - That is repeated

For example:

Input	Result
5 1 2 3 4 1	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[i] == a[j]) {
                printf("%d", a[i]);
                return 0;
            }
        }
    }

    return 0;
}
```

	Input	Expected	Got	
✔	11 36 9 7 6 5 2 8 8 1	8	8	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Find Duplicate in Array

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element a - That is repeated

For example:

Input:	Result:
5 1 3 4 4 5	4

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    int a[n];
    for (int k = 0; k < n; k++)
        scanf("%d", &a[k]);
    for (int k = 0; k < n; k++) {
        for (int j = k + 1; j < n; j++) {
            if (a[k] == a[j]) {
                printf("%d", a[k]);
                return 0;
            }
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 3 4 8 9 1	1	1	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 3 4 4 4	4	4	✔